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ORIGINAL INVESTIGATION

Pooled Versus Individualized Load–Velocity Profiling in the Free-Weight Back Squat and Power Clean

Steve W. Thompson, David Rogerson, Alan Ruddock, Harry G. Banyard, and Andrew Barnes

Purpose: This study compared pooled against individualized load–velocity profiles (LVPs) in the free-weight back squat and power clean. ***Methods:*** A total of 10 competitive weightlifters completed baseline 1-repetition maximum assessments in the back squat and power clean. Three incremental LVPs were completed, separated by 48 to 72 hours. Mean and peak velocity were measured via a linear-position transducer (GymAware). Linear and nonlinear (second-order polynomial) regression models were applied to all pooled and individualized LVP data. A combination of coefficient of variation (CV), intraclass correlation coefficient, typical error of measurement, and limits of agreement assessed between-subject variability and within-subject reliability. Acceptable reliability was defined a priori as intraclass correlation coefficient > .7 and CV < 10%. ***Results:*** Very high to practically perfect inverse relationships were evident in the back squat (r = .83–.96) and power clean (r = .83–.89) for both regression models; however, stronger correlations were observed in the individualized LVPs for both exercises (r = .85–.99). Between-subject variability was moderate to large across all relative loads in the back squat (CV = 8.2%–27.8%) but smaller in the power clean (CV = 4.6%–8.5%). The power clean met our criteria for acceptable reliability across all relative loads; however, the back squat revealed large CVs in loads ≥90% of 1-repetition maximum (13.1%–20.5%). ***Conclusions:*** Evidently, load–velocity characteristics are highly individualized, with acceptable levels of reliability observed in the power clean but not in the back squat (≥90% of 1-repetition maximum). If practitioners want to adopt load–velocity profiling as part of their testing and monitoring procedures, an individualized LVP should be utilized over pooled LVPs.

Keywords: velocity-based training, load–velocity relationship, maximal strength, resistance exercise, strength and conditioning

Training intensity is typically derived from direct assessments (1-repetition maximum [1RM]) followed by relative submaximal load prescriptions (eg, 85% of 1RM).[1] Despite 1RMs showing good within-subject reliability,[1,2] it is hypothesized that this approach might struggle to account for acute changes in maximum strength or residual fatigue build-up.[2] Research has indicated that 1RM can significantly increase following acute bouts of resistance training (1–4 wk).[3–5] Significant decreases in 1RM as a result of residual fatigue (24 h to 1 wk in duration) are also evident,[4,6] potentially affecting the accuracy of prescriptions on a week-to-week basis. Regular 1RM assessments are possible; however, practitioners are faced with time constraints and logistical impracticalities. Such drawbacks have prompted the development of additional aids and approaches to maximal strength testing, such as the load–velocity profile (LVP).

Strong inverse relationships have been observed between load and barbell velocity in free-weight[2,7–9] (r > .93) and fixed-path Smith machine exercises[10–14] (r > .90). However, the application of this method has often been dictated by the procedures employed. For example, the inclusion of Smith machines, pauses between the eccentric and concentric phases of a lift, single-session methodologies, and a failure to investigate the reliability of velocity across a full spectrum of loads brings into question the practical representation of many of these studies to an applied setting in which free-weight and full isotonic exercises are utilized. Furthermore, different modalities of training (eg, Smith machine vs free weights or concentric only vs eccentric–concentric) produce different kinematic outputs and LVPs,[12,15] highlighting the need for further research that investigates the reliability of velocity across a full spectrum of loads during multiple testing sessions in free-weight, full isotonic exercises.

A paucity of research, however, has begun to investigate more practically representative training methods such as free-weight exercises that utilize the stretch-shortening cycle. Banyard et al[2,7] observed high intraclass correlation coefficients (ICCs ≥ .81), low coefficient of variation (CV ≤ 9.1%), and small standard error of measurements (≤0.07 $m \cdot s^{-1}$) between 3 separate LVP trials in loads ≤90% of 1RM and a strong relationship between load and velocity (r ≥ .93) in the free-weight back squat. Similar values were found in the free-weight prone bench pull, bench press, and deadlift.[8–10] Recent data, however, have highlighted that the reliability of LVPs is potentially load dependent[16]; that large between-subject variability at submaximal loads (CVs > 10%) is evident[7,11]; and that poor reliability of velocity at 1RM (ICC = .19–.66; CV = 15.7%–22.5%) can also be observed across a range of exercises.[2,7–10] Moreover, individualized LVPs seemingly provide stronger relationships between load and velocity.[7,10,11] With clear uncertainties about the most effective way to construct LVPs, further research in free-weight exercises investigating the individuality of load–velocity characteristics is needed.

The LVPs are traditionally fitted with either linear regression[7] or nonlinear equivalents, such as second-order polynomials.[13,14] A small number of studies have compared the 2 statistical models[2,8,10]; however, these have often been limited to Smith machine or upper body exercises. Nevertheless, Banyard et al[2] did investigate this comparison during the free-weight back squat and found no statistical differences; however, the small number of loads (6) used to construct the LVP may account for this. Therefore, further

Thompson, Rogerson, Ruddock, and Barnes are with the Dept for Sport and Physical Activity, Sheffield Hallam University, Sheffield, United Kingdom. Banyard is with the Dept of Health Science, Swinburne University of Technology, Melbourne, VIC, Australia. Thompson (s.w.thompson@shu.ac.uk) is corresponding author.

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clarification is required to assess the most appropriate statistical model to apply when constructing a full LVP (>6 loads and <20% increments). Further investigation is also needed into the strength of the load–velocity relationship when employing free-weight isotonic exercises, constructing the profile individually and when utilizing more explosive movements such as weightlifting derivatives.

Weightlifting derivatives such as the power clean are common in strength and conditioning (S&C) interventions, because they train important movement patterns such as the triple extension[17] and are strongly linked to physical characteristics such as sprinting and jumping.[18] Weightlifting stimulates high levels of force generation, rate of force development, and impulse,[17,19] requiring greater acceleration of heavier loads in comparison to biomechanically similar exercises such as loaded squat jumps.[20] High levels of intersession and intrasession reliability in experienced, novice, and youth lifters (ICC > .98; typical error of measurement (TE) = 2.9 kg and smallest detectable differences = 3.76 kg)[19,21,22] have also been reported when performing this exercise incrementally to 1RM. The explosive nature of the power clean and the technical competency required to perform this lift might impact load–velocity characteristics. The margin for error in successfully executing this exercise, therefore, may be smaller than the margin for error in the back squat, and it is proposed that heavier relative loads are likely to be performed at faster velocities and in smaller increments. Importantly, limited research is available that fully assesses LVPs in the power clean. Naclerio and Larumbe-Zabala[23] investigated the LVP in this exercise but only measured peak velocity (PV) and did not assess reliability or evaluate the most appropriate method for constructing the profile. Moreover, our study is the first to evaluate these important considerations when seeking to implement LVP in weightlifting exercises.

Therefore, the primary aim of this study was to investigate the load–velocity relationship of the free-weight back squat and power clean exercises, comparing pooled versus individualized LVPs and linear versus nonlinear regression models. Secondary aims were to determine between-subject variability and within-subject reliability at each relative load for both exercises.

Methods

Design

A repeated-measure within-subject design investigated the reliability of pooled (all subject data combined) and individualized (1 profile for 1 subject) LVPs in the free-weight back squat and power clean. Assessments of 1RM were conducted for each exercise, followed by 3 incremental LVPs utilizing loads of 30%, (back squat only), 40% to 80% (in 10% increments), and 85% to 100% (in 5% increments), with mean velocity (MV) and PV recorded for each repetition.

Subjects

A total of 10 (8 male and 2 female) healthy competitive weightlifters (age = 25.0 [5.6] y; body mass = 73.6 [13.9] kg; stature: 169.6 [6.6] cm) who had competed at a minimum of regional level within the previous 12 months and possessed appropriate relative strength levels (squat > 1.5 × body mass and power clean > 1.15 × body mass) were recruited. Subjects' relative (absolute) strength values were: 2.1 (0.3) (157.0 [35.8] kg) and 1.4 (0.2) (104.4 [22.8] kg) for the back squat and power clean, respectively. Informed consent was provided prior to data collection with ethical approval granted by Sheffield Hallam University's ethics committee in accordance with the seventh revision (2013) of the Declaration of Helsinki.

Methodology

Subjects attended 4 sessions, each separated by 48 to 72 hours. Each session occurred at the same time of day, and participants were asked to perform no additional exercise during data collection. Body mass (in kilograms, InBody 720; Biospace, Seoul, Korea), stature (in centimeters, Harpenden; Holtain Ltd, Wales, United Kingdom), and rack height (in centimeters) were all recorded during the initial visit. Subjects undertook a standardized individualized warm-up that included 5 minutes on a cycle ergometer (Ergomedic 874 E; Monark Exercise AB, Monark, Vansbro, Sweden) at 100 W followed by a combination of body weight movements, mobility exercises, and light barbell lifts. Baseline 1RM assessments were then conducted in the power clean (AM) followed by the back squat (PM). A calibrated International Weightlifting Federation–approved 20-kg Olympic barbell and bumper plates (Werksan, Akyurt, Turkey) and portable squat rack (Mirafit, Norfolk, United Kingdom) were used throughout the study. The 1RM protocols started at an estimated 50% of 1RM and increased incrementally until 1RM was reached. Multiple repetitions were performed at warm-up loads (5 repetitions at 50% of 1RM; 3 repetitions at 70% and 80% of 1RM) with single repetitions for all remaining loads (85%, 90%, 95%, and 100% of 1RM). Up to 5 attempts were allowed to determine a true 1RM, with loads being increased by 0.5 to 5 kg. Rest periods were 3 to 5 minutes between all sets. Subjects were habituated to performing lighter loads with maximal intent and velocity during this visit.

The 3 subsequent LVP sessions were identical in procedure and consisted of incremental protocols for the power clean and then the back squat; loads were determined from baseline 1RM. Three repetitions were performed for lighter loads (30%–60% of 1RM), 2 repetitions for moderate loads (70% and 80% of 1RM), and 1 repetition for heavy loads (85%–100% of 1RM). Up to 5 attempts were permitted to achieve the 100% of 1RM load. Rest periods were 3 to 5 minutes between all sets.

For all successful repetitions of the power clean and back squat, techniques met the International Weightlifting Federation and International Powerlifting Federation guidelines and was in accordance with previous research.[2,17,21,24,25] A power clean was deemed successful if, upon catch, the greater trochanter of the hip was superior to the lateral epicondyle of the knee and the subject was able to fully extend the lower limbs.[17,21] The back squat required subjects to descend, ensuring the greater trochanter was inferior to the lateral epicondyle of the knee at full descent and the subject could fully extend the lower limbs on ascent.[2,24] Technical competency of both exercises was evaluated via a simple 2d video assessment (iPhone 7, version iOS 11.4.1; Apple, Cupertino, CA) and an experienced S&C coach. Subjects were instructed to perform the ascents of both lifts as quickly and explosively as possible for all loads and to perform the descent at a natural speed.

GymAware was used to measure MV and PV during each repetition; it has previously been shown to be reliable and valid in measuring barbell velocity.[26] MV refers to the velocity recorded across the full concentric phase of the lift (propulsive and braking phases), with PV referring to the instantaneous maximum velocity recorded during the concentric phase. The tether of the device was attached to the right-hand collar of the barbell, 100 mm from the end of the bar. The unit was placed directly under the bar for each repetition, with a tether angle of 0° (5°).

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Statistical Analysis

Normal distribution and relevant assumptions were assessed prior to analysis. Linear and nonlinear (second-order polynomial) regression models were fitted to the pooled and individualized data in order to assess the relationship between load and MV or PV. Fisher *r*- to *z*-transformations were used to determine significant differences between linear versus nonlinear regression model correlation coefficients.[2]

Pearson product–moment correlations (*r*) and standard error of the estimate (SEE) assessed the relationship between load and velocity. The strength of the correlations was determined using the following criteria: trivial (<0.1), small (0.1–0.3), moderate (0.3–0.5), high (0.5–0.7), very high (0.7–0.9), or practically perfect (>0.9).[27] Between-subject variability at each relative load was analyzed using CV (CV [%] $= \frac{\text{Between–subject SD}}{\text{subject mean score}} \times 100$). Within-subject reliability at each relative load was assessed using ICC (model 3.1), CV (CV [%] $= \frac{\text{Within–subject SD}}{\text{subject mean score}} \times 100$), TE, and Bland–Altman limits of agreement (LOA) (95% confidence interval). Within-subject reliability refers to the reliability between sessions. The reliability of the 1RM data was assessed via 1-way repeated-measures analysis of variance, partial eta-squared effect sizes (η_p^2), ICC, CV, and TE. All 3 trials were used for all reliability analyses except for LOA. For LOA, trials 1 and 3 were utilized in order to allow for the largest impact of habituation and residual fatigue on the data. Statistical significance was set at $P < .05$ for all relevant statistical tests. Magnitudes of the CVs were determined as: large (>10%), moderate (5%–10%), and small (<5%).[7] Acceptable reliability was defined a priori as very high correlation (>.70) and small to moderate CV (<10%).[2] Smallest worthwhile change (SWC) was calculated for each relative load of both exercises.

Results

Data were normally distributed and met the assumptions for regression. A very high to practically perfect inverse relationship was found between velocity and load for both exercises (Figure 1, Table 1). The group's maximum load (in kilograms) during each LVP session demonstrated an acceptable level of reliability in the back squat ($P = .17$; $\eta_p^2 = .18$; ICC = .99; CV = 1.8%; TE = 2.69 kg) and power clean ($P = .99$; $\eta_p^2 = .001$; ICC = .99; CV = 2.0%; TE = 1.84 kg), indicating true 1RMs were observed in each session and confounding variables such as residual fatigue were controlled for.

Linear regression and second-order polynomials were fitted to the pooled LVPs of the sample and indicated very strong to practically perfect relationships between load and velocity for the back squat and power clean (Table 1). Individualized LVPs were then analyzed using the same approaches. Individualized LVPs

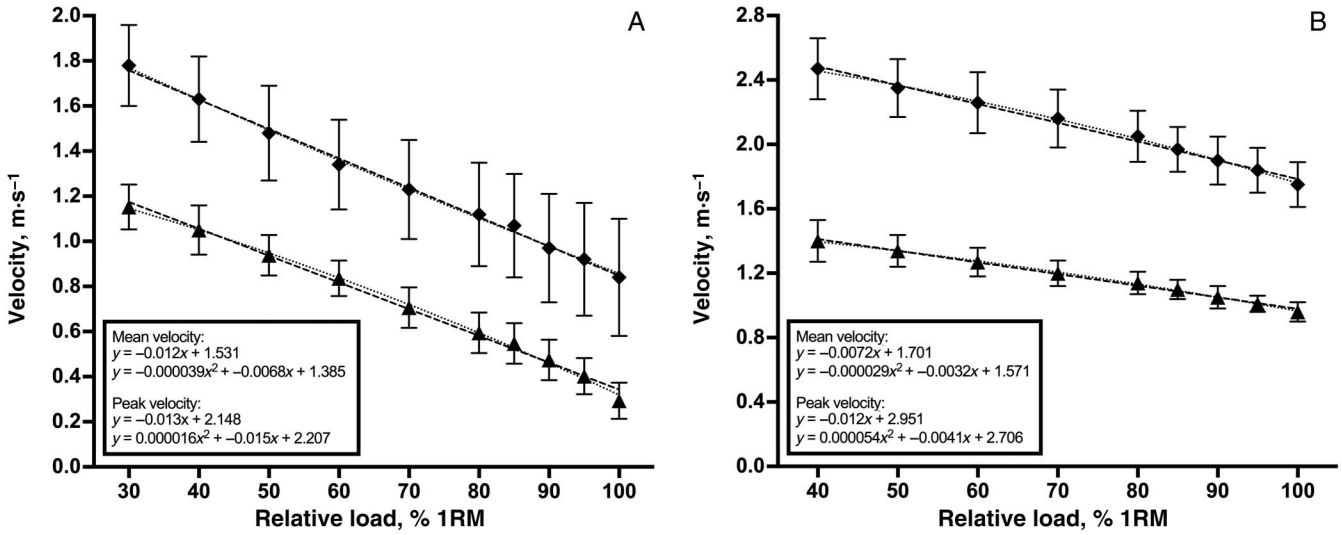

Figure 1 — Group mean (SD) values from 3 load–velocity profiles for MV (m·s^{-1}) (▲) and PV (m·s^{-1}) (◆) for (A) back squat and (B) power clean. Linear regression (- - -) and second-order polynomial (····) are presented with respective equations (located in box). 1RM indicates one-repetition maximum; MV, mean velocity; PV, peak velocity.

Table 1 Linear Regression and Second-Order Polynomials Correlation Coefficients (*r*) With SEE for the Back Squat and Power Clean, Pooled Versus Individualized Data

		Linear regression				Second-order polynomial			
		Pooled		Individualized		Pooled		Individualized	
Exercise		*r*	SEE, m·s^{-1}	*r*	SEE, m·s^{-1}	*r*	SEE, m·s^{-1}	*r*	SEE, m·s^{-1}
Back squat	MV	.96	0.09	.98–.99	0.02 to 0.06	.96	0.09	.98–.99	0.02 to 0.05
	PV	.83	0.22	.96–.99	0.03 to 0.11	.83	0.22	.98–.99	0.01 to 0.05
Power clean	MV	.89	0.08	.87–.99	0.02 to 0.06	.90	0.08	.92–.99	0.01 to 0.04
	PV	.83	0.16	.85–.99	0.02 to 0.10	.83	0.16	.85–.99	0.01 to 0.09

Abbreviations: MV, mean velocity; PV, peak velocity; SEE, standard error of estimates.

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were stronger for all data sets but were substantially stronger for PV in both lifts (Table 1). All correlations were statistically significant ($P = .001$). Fisher r- to z-transformations revealed no significant differences (back squat: $P = .45$; power clean: $P = .50$) between the linear and nonlinear regression models (Table 1). Large CVs for between-subject variability were present in the back squat (>10%) for a number of relative intensities for MV (70%–100% of 1RM) and PV (40%–100% of 1RM) (Figure 2). The power clean presented CVs < 10% for all relative loads (Figure 3).

The systematic bias and LOAs (95%) between trials 1 and 3 were as follows: 0.009 (0.06) $m \cdot s^{-1}$ (MV) and –0.002 (0.14) $m \cdot s^{-1}$ (PV) for the back squat and 0.001 (0.05) $m \cdot s^{-1}$ (MV) and 0.004 (0.07) $m \cdot s^{-1}$ (PV) for the power clean (Figure 4). Within-subject reliability can be seen in Figures 5 and 6. MV and PV presented ICCs of .82 to .98, CVs of 2.1% to 4.9%, and TEs of 0.03 to 0.07 $m \cdot s^{-1}$ for all relative intensities in the power clean, meeting the criteria for acceptable reliability. The back squat, however, did not meet the criteria for acceptable reliability at relative intensities of ≥95% (ICC = .75–.86; CV = 13.1%–20.6%; TE = 0.03–0.06 $m \cdot s^{-1}$) and ≥90% (ICC = .87–.91; CV = 11.8%–15.6%; TE = 0.10–0.14 $m \cdot s^{-1}$) for MV and PV, respectively. MV and PV SWC for each relative load for both exercises can be seen in Table 2.

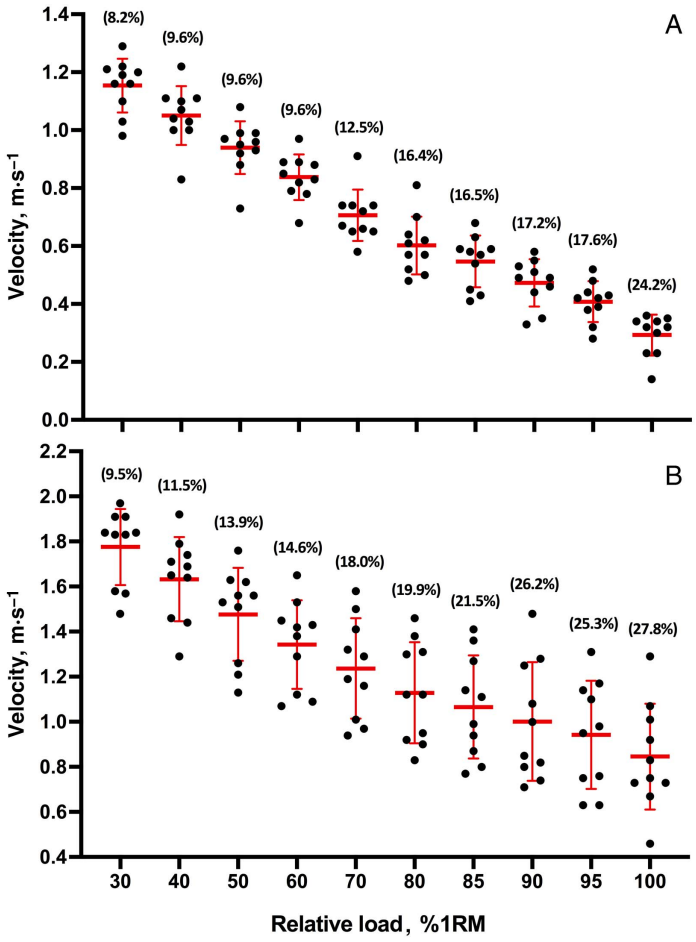

Figure 2 — Between-subject variability for MV ($m \cdot s^{-1}$) (A) and PV ($m \cdot s^{-1}$) (B) for the back squat. Mean (SD) are represented by the horizontal bar (error bars). CV is displayed above each relative load in parentheses. 1RM indicates one-repetition maximum; CV, coefficient of variation; MV, mean velocity; PV, peak velocity.

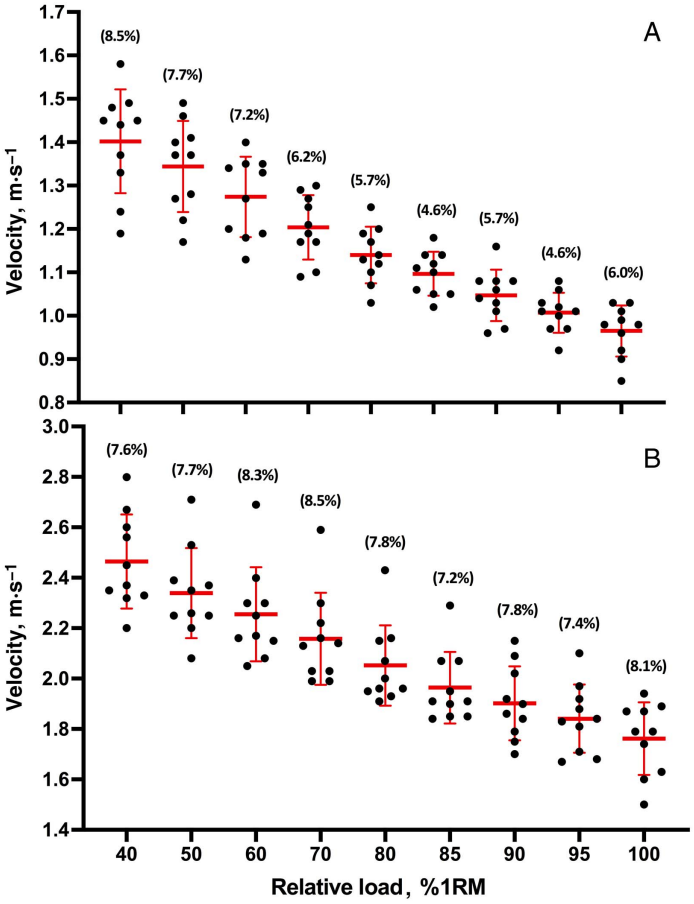

Figure 3 — Between-subject variability for MV ($m \cdot s^{-1}$) (A) and PV ($m \cdot s^{-1}$) (B) for the power clean. Mean (SD) are represented by the horizontal bar (error bars). CV displayed above each relative load in parentheses. 1RM indicates 1-repetition maximum; CV, coefficient of variation; MV, mean velocity; PV, peak velocity.

Discussion

The primary aim of this study was to investigate the load–velocity relationship of the free-weight back squat and power clean exercises, comparing pooled versus individualized LVPs and linear versus nonlinear regression models. The primary findings of this investigation were as follows: (1) the back squat and power clean demonstrated strong inverse relationships between load and velocity, with stronger relationships observed from individualized LVPs and no statistical differences observed between the 2 regression models; and (2) the back squat demonstrated moderate-to-large between-subject variability, whereas the power clean displayed much lower variability.

Very high to practically perfect inverse relationships ($r = .81–.96$) were observed between load and velocity for both exercises (Figure 1 and Table 1), reflecting existing data in the free-weight back squat (r and $R^2 = .93–.99$).[2,7] The impact of crossbridge cycling on force production is thought to underpin this association. As the shortening of a muscle quickens, actin and myosin have less time for crossbridges to form, inhibiting force production.[28] Comparable studies for the power clean are scarce; however, it is evident that the LVP of the power clean is unique (Figure 1), indicating that load–velocity relationships are exercise specific. Naclerio and Larumbe-Zabala[23] suggested that only 46%

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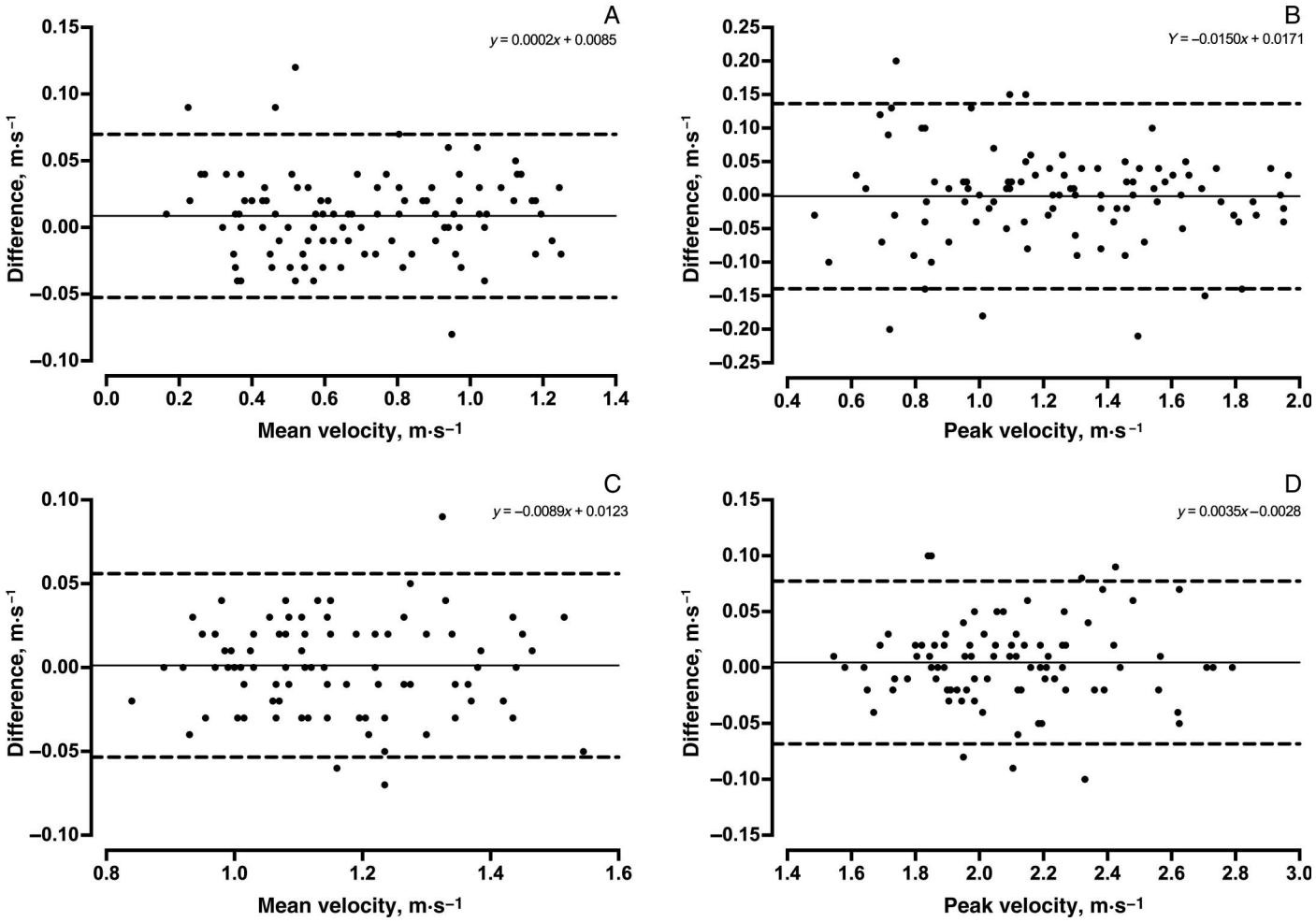

Figure 4 — Bland–Altman plots exhibiting variations in MV ($m \cdot s^{-1}$) (A and C) and PV ($m \cdot s^{-1}$) (B and D) between trials 1 and 3 measured in 10% increments (30%–80% of 1RM) and 5% increments (85%–100% of 1RM) for the back squat (A and B) (n = 100) and 10% increments (40%–80% of 1RM) and 5% increments (85%–100% of 1RM) for the power clean (n = 90) (C and D). — represents mean systematic bias and - - - represents LOA (95% confidence intervals). 1RM indicates 1-repetition maximum; LOA, limits of agreement; MV, mean velocity; PV, peak velocity.

of variance could be explained when using PV to predict relative load (percentage of 1RM). This suggests a much lower correlation compared with our data, potentially due to the technical competency of the elite sample recruited for the present study. Similarly, comparisons to MV with Naclerio's data are not possible, limiting the interpretation of their research. Furthermore, the application of the LVP when applied to the power clean may differ depending on the velocity characteristic of interest. PV is most likely to occur during the second pull phase,[17] providing greater insight into an individual's explosive strength, whereas MV may be a more stable metric to monitor and is largely determined from the first pull and transition phases.

The authors observed large between-subject variability across relative loads in the back-squat exercise, with CVs of up to 24.2% and 27.8% for MV and PV, respectively (Figure 2). This finding is similar to that of Balsalobre-Fernandez et al,[11] who observed CVs of up to 24.6% when performing a seated military press in a Smith machine, and Banyard et al,[2] who reported large absolute differences between subjects across all loads (0.33–0.68 $m \cdot s^{-1}$) in the free-weight back squat. This variability could be a contributing factor to the poor application of predetermined generalized predictive equations such as those developed by González-Badillo and Sánchez-Medina.[13] Garcia-Ramos et al[29] investigated the use of these predictive equations to estimate 1RM and observed large discrepancies from the measured maximal loads (2.8–11.4 kg) when using MV. Furthermore, better results were obtained when employing an individualized LVP (0.6–2.6 kg). Research has shown that individuals with similar 1RM values may produce different force–velocity profiles depending on their neuromuscular properties, such as fiber type, recruitment patterns, and synergistic coordination,[1,28,30,31] highlighting the need to profile athletes individually. This may facilitate the development of individualized training programs as well as optimize the efficiency and effectiveness of a training intervention to elicit desired training effects.

The between-subject variability within the power clean was lower than that of the back squat (CVs of <10%; Figure 3). Again, stronger correlations were found for an individualized LVP than for the pooled profiles (Table 1). Further, within-subject variability (CVs—Figure 6) was lower than between-subject variability (CVs—Figure 3) across all relative loads, indicating that individualized LVPs are favorable. This relationship has previously been reported for the bench press and prone bench pull,[8,10,32] similar to our data and indicating that individualized LVPs are a more accurate and reliable measurement when training and testing athletes.

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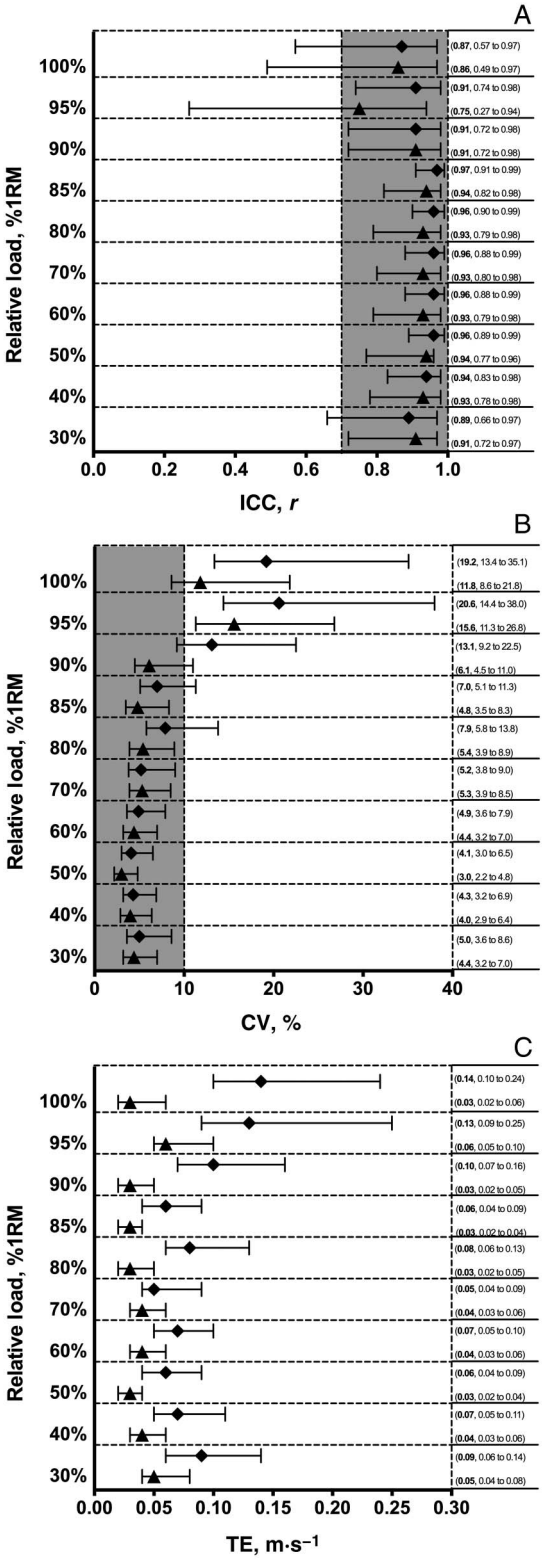

Figure 5 — Within-subject reliability of MV ($m \cdot s^{-1}$) (▲) and PV ($m \cdot s^{-1}$) (⧫) in the back squat at all submaximal relative loads. Forest plots displaying ICCs (A), CV (B), and TE (C) with error bars indicating 95% confidence intervals. Right *y*-axis details group mean and 95% confidence values. Gray-shaded areas indicate the criteria for acceptable reliability defined a priori. 1RM indicates 1-repetition maximum; CV, coefficient of variation; ICCs, intraclass correlations coefficients; MV, mean velocity; PV, peak velocity; TE, typical error.

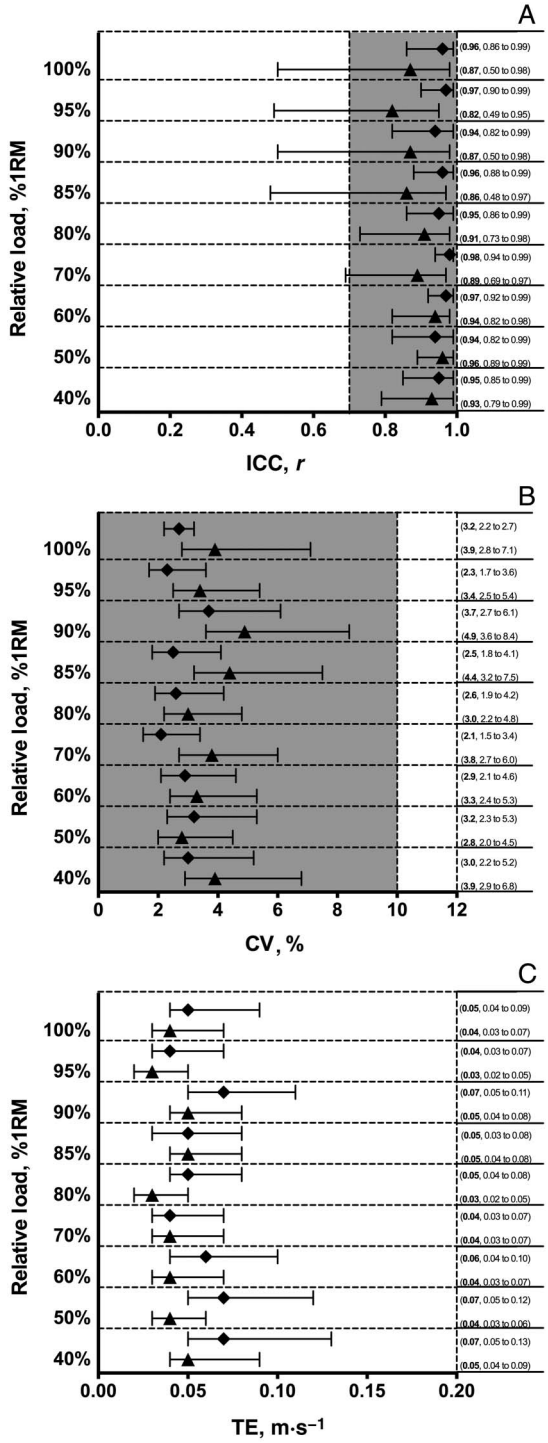

Figure 6 — Within-subject reliability of MV ($m \cdot s^{-1}$) (▲) and PV ($m \cdot s^{-1}$) (⧫) in the power clean at all submaximal relative loads. Forest plots displaying ICCs (A), CV (B), and TE (C) with error bars indicating 95% confidence intervals. Right *y*-axis details group mean and 95% confidence values. Gray-shaded areas indicate the criteria for acceptable reliability defined a priori. 1RM indicates 1-repetition maximum; CV, coefficient of variation; ICCs, intraclass correlations coefficients; MV, mean velocity; PV, peak velocity; TE, typical error.

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Table 2 Recommendations for the SWC of MV and PV for Each Relative Load Performed Across Both Exercises

Load (% of 1RM)	Back squat		Power clean	
	MV, m·s^{-1}	PV, m·s^{-1}	MV, m·s^{-1}	PV, m·s^{-1}
30	0.02	0.04		
40	0.02	0.04	0.03	0.04
50	0.02	0.04	0.02	0.04
60	0.02	0.04	0.02	0.04
70	0.02	0.04	0.02	0.04
80	0.02	0.05	0.01	0.03
85	0.02	0.05	0.01	0.03
90	0.02	0.05	0.01	0.03
95	0.02	0.05	0.01	0.03
100	0.02	0.05	0.01	0.03

Abbreviations: 1RM, 1-repetition maximum; MV, mean velocity; PV, peak velocity; SWC, smallest worthwhile change.

Both exercises in this study exhibited strong inverse relationships (Figure 1). The use of nonlinear regression models (second-order polynomials) has been proposed as a method of strengthening the predictive model.[2,10] Our data supports that of previous research showing that no statistical differences are evident between the 2 regression models in either exercise ($P > .05$; Table 1).[2,10] Therefore, either approach could be implemented depending on the preference of the practitioner and the number of loads included in the profile.

The secondary aim of this study was to determine the within-subject reliability of the LVPs and velocity measures at each relative load. To our knowledge, this is the first study to examine the between-session reliability of load–velocity profiling in the power clean. Importantly, the authors observed high repeatability in the 1RM data (in kg) across the 3 sessions in both exercises, indicating that 1RM testing is a reliable method for assessing maximal strength as well as demonstrating the robustness of our methodology. Despite this, previous research has indicated that 1RM can significantly change with respect to strength developments and fatigue build up over a short-time period[3–6]; therefore, frequent 1RM assessments to monitor changes in strength are not always desirable, particularly during in-season competition.

When evaluating LVPs as a whole, the authors observed minimal systematic bias between trials in both exercises (–0.002 to 0.009 m·s^{-1}), with 95% confidence intervals of 0.05 to 0.06 and 0.07 to 0.14 m·s^{-1} for MV and PV, respectively (Figure 4). Given the scale of the unit of measure, the 95% confidence intervals could indicate important methodological considerations. For example, accurate manipulation of load could be compromised if the associated measurement error is not taken into account by practitioners. The SWC (Table 2) provides practitioners with practical values that represent meaningful changes. By using this data, coaches can be more confident that improvements to load-velocity data have occurred as a result of effective training strategies as opposed to measurement or procedural error. The smaller SWCs observed for PV in the present study compared with MV suggests that MV is perhaps the better metric to use in order to evaluate the effectiveness of training interventions.

Analyzing LVPs as a whole could limit their practical use given that prescriptions typically occur from specific relative loads (eg, 85% of 1RM). The power clean produced acceptable levels of reliability across all relative loads in MV and PV (Figure 6), suggesting that it could be utilized as an appropriate tool for practitioners to test and monitor the progress of their athletes. Conversely, the back squat did not meet the reliability criteria for loads ≥95% for MV and ≥90% for PV (CVs = 13.1%–20.6%) (Figure 5). This was in agreement with previous research that observed moderate ICCs (.55–.63) and large CVs (15.7%–19.4%) at heavier loads (>90%) when measuring velocity in the free-weight back squat and deadlift.[2,7,9] However, practitioners could look to utilize LVPs of 30% to 90% of 1RM using MV, given the low to moderate CVs and TEs (3.0%–6.1% and 0.03 to 0.05 m·s^{-1}, respectively; Figure 5).

Small horizontal movements and the influence of the stretch-shortening cycle have previously been attributed to poorer within-subject reliability at heavy loads.[2,7,9] Furthermore, biomechanical deviations may affect the path of the barbell, altering kinematic variables such as barbell velocity. For example, significant inter-individual and intraindividual variability in barbell velocity and hip, knee, and ankle angular velocity in a 90% of 1RM back squat have previously been reported.[33] The better within-subject reliability in the power clean observed in our study further reinforces this argument. The power clean is technically more complex, requiring faster velocities in order to successfully complete a lift (Figure 1). This smaller margin for error requires greater consistency in the biomechanical positioning achieved from repetition to repetition. For example, differences of ≥8 cm in forward barbell displacement, ≤0.19 m·s^{-1} in barbell velocity, and ≤33° resultant acceleration angle in the second-pull phase can dictate the success of a repetition.[17] Therefore, movement variability may contribute to the poorer reliability evident at heavier loads in the back squat.

Despite favorable reliability data for the LVP, a full-individualized profile, if performed in a similar way to the present study, may still be time consuming and logistically difficult. Furthermore, if adopting such a method, it is advised that practitioners aim to do so alongside more traditional 1RM testing, given the acceptable reliability of the 1RM data observed in this study when free from confounding variables. This combination will ensure that S&C coaches are able to accurately and reliably measure the maximum strength capabilities of their athletes (1RM) and optimally manipulate load session-to-session (LVP). Practitioners, however, must be cognizant of the limitations that surround the construction, application, and utilization of LVPs if opting to employ them in their practices.

Practical Applications

S&C practitioners wanting to profile an athlete's load–velocity characteristics should ensure that an individualized approach is utilized. Practitioners should evaluate the need for profiling their athletes, the time and equipment available, and factor in the SWC associated with each relative load. S&C coaches should not replace traditional methods such as the 1RM with LVPs, but should instead consider the addition of LVPs in order to assist in testing and monitoring. For example, warm-up sets of an incremental protocol utilized during a 1RM assessment could be used to form the light to moderate loads of an LVP. Despite this, practitioners should be cognizant of the logistical and time-related issues surrounding individualized LVPs and should adopt a method that will fit into the scope of their practices. Finally, if undertaking LVPs in the free-weight back squat, practitioners should be mindful of the associated error when performing this method multiple times and adjust the approach accordingly.

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Conclusions

Load and velocity demonstrate a very strong to practically perfect inverse relationship in the free-weight back squat and power clean. However, large between-subject variability or a smaller within-subject to between-subject variability ratio, indicates that load–velocity characteristics are highly individualized. The back squat evidenced poor within-subject reliability in MV and PV during the heavier loads (≥90% of 1RM), perhaps due to greater movement variability; however, MV and PV demonstrated high within-subject reliability across all relative loads in the power clean.

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